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INTERNET OF THINGS

The internet of things (IOT) refers to uniquely identifiable objects and their virtual representations in an internet like structure.

ITU — International Telecommunication Union

ITU Definition

The IOT can be viewed as a global infrastructure for the information society, enabling advance services

Things are objects of the physical world (physical things) or the information world (virtual world) which are capable of being identified and integrated into communication network

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Market Segment Example, IoT Value Stack and Technology

Internet of things IoT refers to the networked connection of physical object (things) through the use of sensors, actuators and other devices that are capable of sensing or acting on their

IoT Segmentation

All systems are identifiable, predictable, operational patterns. We define behaviour a little differently when talking about consciousness (human and otherwise, but behavior is defined by predictable patterns at its core.

The segmentation of IoT uses two important set of behavioral tensions

1. Needs; Survive Vs. Thrive: This tension is between system that

ensure that we live and systems that help us improve how we live and help us to live well.

2- Input: Autonomous Vs. Reactive.
Another way of expressing this tension would be self-directed Vs. interactive or responsive behaviour

IoT Market segment

- Monitoring segment: focusing on sensing condition, recognizing patterns and reporting abnormalities (Eg. security sensors, utility monitoring and remote diagnostic
- Operational segment: This include system that perform high-assurance, precision, repetitive tasks such as robotics, industrial machines and automated manufacturing lines.
- Life segment: Major appliances like refrigerator, residential air conditioner from manufacturers like Sharp and Bosch automate more fundamental

human needs.

- Attainment segment : Enhances life style quality with aspirational technologies such as wearable, home automation systems and entertainment IOT devices

SEGMENT	Human Centric Expectation	IOT system Expectation
Monitoring	Effectively manage repetitive services	Recognize patterns

MAJOR IOT MARKET SEGMENT

1. Consumer IOT [smart tv, smart locks]
~~Smartphones~~ Smart home devices
Wearables
Home automation

2. Industrial IOT [siemens automation]
Robotics
predictive maintenance
asset monitoring

3. Health care IOT [Heart rate monitors, glucose sensors]

- Remote patient monitoring
- smart hospital equipment

4 Smart Cities [IoT streetlight]
- smart traffic light
- Waste management
- smart grids

5. Automotive IoT [Tesla's sensor]
- connected cars
- Fleet monitoring
- ADAS system

6 Agriculture IoT [Moisture sensors]
- Drone crop monitoring

VALUE CHAIN

Upstream — space manufacturers

Middlestream — Connectivity service providers
IoT service providers

Downstream → Hardware manufacturers
system integrators
Application providers

Users → End Users

WIRELESS COMMUNICATION TECHNOLOGIES

1. Short range networks for localised connectivity and
2. Wide area network to connect wireless devices over large areas.

These two broad types of connectivity technologies and can be sub-divided further as follows:

- Short range wireless communication Technology: They include local area network (LAN) and personal area network (PAN such as Bluetooth and ZigBee
- Wide area wireless communication Technology: They include cellular network, Low Power Wide Area (LPWA) Two of the main providers of LPWAN are

LoRa (Long ~~area~~ range radio)
Sigfox

Wireless Communication Requirement of IOT / M2M

1. Range – The extent to which an application favour short range or wide area technology
2. Bandwidth : The extent to which an application requires the transmission of a large amount of data at a high frequency and rate (wide band) or not (narrow band).
3. Quality of Services (QOS) : The extent to which an application is mission-critical and therefore in need of a connection with a high degree of security and reliability.

THE IOT LAYERS / MODEL

7 Collaboration & processes
6. Application (Reporting, Analytics & Control)
5 Data Abstraction (Aggregation & Access)
4. Data Accumulation (storage)

3 Edge computing
2- Connectivity
1. physical devices & Controllers